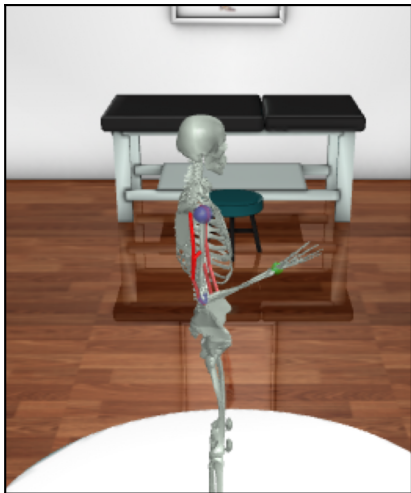


ReGainX Three-Phase Training Pipeline

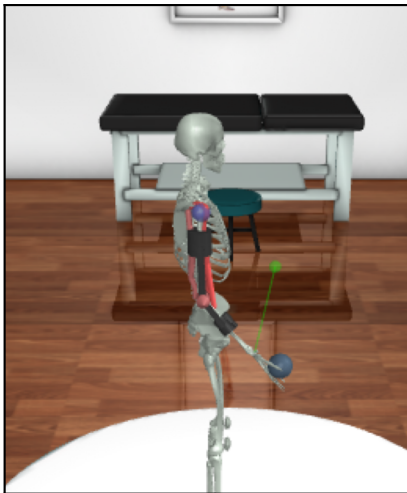
target = 2.0 rad

Phase 1: Healthy Controller



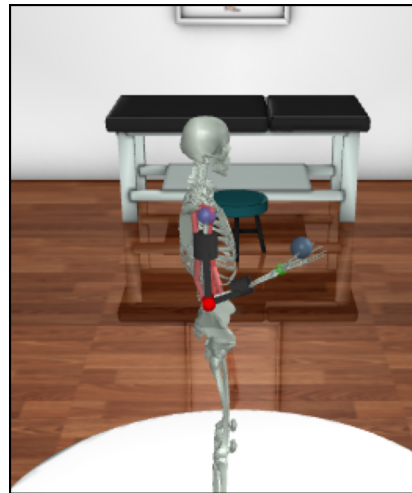
PPO trains the 6-muscle elbow (myoElbow, 250k steps). The healthy arm reaches the 2.0 rad target.

Phase 2: PD Patient (Frozen)



Healthy policy is frozen and run under bradykinesia and muscular degeneration. It stalls at 0.7 rad; the target is missed.

Phase 3: Exoskeleton Agent



RecurrentPPO (LSTM) adds torque to the frozen patient (2M steps, POMDP). Assisted arm reaches the 2.0 rad target.